B.E. I EXAMINATION -MAY' 2024
IT – A ,B and CIVIL
2MER8C3
ENGINEERING DRAWING

Duration: 3 hrs. **Max. Marks: 60**

Note: All questions are compulsory. Attempt any two parts from Question 1 to 3. Assume suitable data wherever if necessary & mention it, Drawing should be Neat.

Q.1. (a) Construct a Vernier scale of 1:40 to read metres, decimetres and centimetres and long enough to measure up to 6 m. Mark a distance of 5.76 m on it.(use both forward and backward Vernier scale). **06**

(b) Draw hypocycloid generated by a rolling circle of 60 mm diameter for one complete revolution .The radius of the directing circle is 100 mm .draw a tangent and a normal to the hypocycloid at 50 mm from the centre of the directing circle . **06**

(c) The major axis of an ellipse is 110 mm long and the foci are at a distance of 15 mm from its ends. Draw the ellipse, one –half of it by 'concentric circles' method and the other half by 'rectangle' method .Determine eccentricity of the ellipse. **06**

Q.2. (a) The front view and top view of a straight line PQ measures 50 mm and 65 mm respectively. The point P is in the H.P. and 20 mm in front of the V.P. and the front view of the line is inclined at 45^0 to the reference line .Determine the true length of PQ, true angles of inclination with the reference planes and the trace. **06**

(b) A thin circular plate with a 60 mm diameter, appears in the front view as an ellipse of major and minor axes, 60 mm and 40 mm in length, respectively. Draw its projections when one of the diameters is parallel to both the reference planes. **06**

(c) A 100 mm long line PQ is inclined at 30° to the H.P. and 45° to the V.P. Its mid point is 35 mm above the H.P. and 50 mm in front of V.P. Draw its projections. **06**

Q.3. (a) Explain the essential difference between the I^{st} angle projection and III^{rd} angle projection. **06**

(b) Draw Front, Side and Top View of the object Shown in ***Figure 1*** *(all dimensions are in mm)* with respect to direction of Front view (X) by First Angle Projection Method. **06**

(c) Draw an isometric view of a circular lamina with a 40 mm diameter on all the horizontal plane and vertical plane using four centre method. **06**

Q.4. A cone, with a 60 mm base diameter and a 75 mm long axis, is resting on its base on H.P. A profile section plane 10 mm away from the axis cuts the cone .Draw the development of lateral surface of retained cone. *12*

OR

A cylinder, with a 50 mm base diameter and 60 mm long axis ,is resting on its base on the H.P. it is cut by a section plane perpendicular to the V.P. , the V.T of which cuts the axis at a point distance from the bottom face and makes an angle of 45^0 with the reference line . Draw its front view, sectional top view and true shape of the section. *12*

Q.5. A hexagonal prism with a 40 mm base side and 100 mm long axis , is resting on its base on H.P. with a side of base parallel to V.P. It is penetrated by a horizontal cylinder with a 50 mm base diameter and 100 mm long axis such that their axes bisect each other at right angles. Draw the projections of the combination and show curves of intersection. *12*

OR

A square prism having a base with a 70 mm side is resting on its base on the H. P. with a face inclined at 30^0 to V. P. It is completely penetrated by a horizontal cylinder with a 70 mm base diameter such that axes of the prism and the cylinder intersect each other at right angles. Draw the projections of the combination and show the curves of intersection. *12*

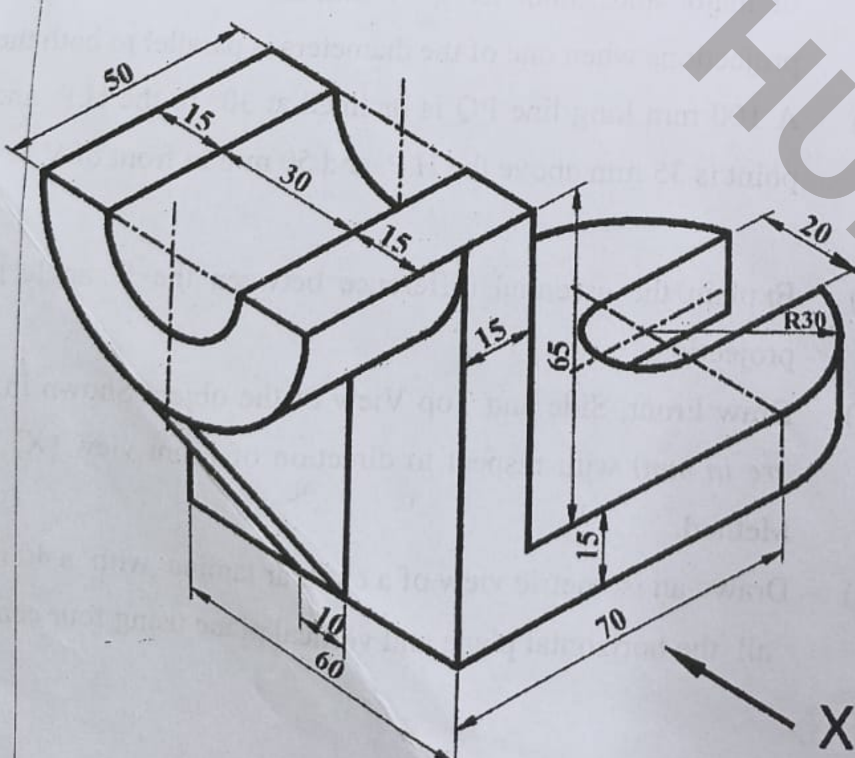

Figure 1